

nabta  WELLNESS PARK

THE LIVING SHADE LOOP

DUBAI MUNICIPALITY • AI PARK DESIGN CHALLENGE • DRAFT COMPETITION ENTRY

# D•08

## Accessibility, Circulation & Commercial Facilities

W08 Accessibility

CSF Map



# Deliverable 8 — Accessibility & Circulation Strategy + Indicative Commercial & Service Facilities Plan

## A. CIRCULATION HIERARCHY — MOVEMENT FOR EVERY BODY

Circulation *is* the programme: walking, wheeling and jogging are Nabta's primary health offer. Across the 15,001 m² parcel — an irregular 16-vertex as-built boundary, roughly 120 × 178 m, street edge to the south — one legible spine and two subordinate layers let every visitor complete the circuit with dignity.

- **Primary: the Living Shade Loop.** An 858 m continuous network — a 398 m figure-eight threaded through the garden rooms plus a 460 m perimeter ring (~1 km including the wadi spur) — on a 3.5 m shared path, continuously shaded. A 1.0 m jogger lane is marked in terracotta-pigmented resin on the outer edge; cross-slope ≤2% throughout, gradients ≤1:21, no ramps needed on any path.
- **Secondary: garden paths.** 1.8 m stabilised, firm and slip-resistant paths entering each of the ten garden rooms, with 1.8×1.8 m passing/turning bays at every junction.
- **Tertiary: desire-line shortcuts.** Compacted decomposed-granite links across the loop's waist and lawn edges; all step-free.

**Walk times** (1.0 m/s accessible pace): entry plaza → play dune 3 min; → sensory wadi 5 min; full 858 m network 14 min walking, 6 min jogging. No point is more than 4 minutes from an exit gate.

## B. UNIVERSAL ACCESSIBILITY — CODE-MAPPED CHECKLIST

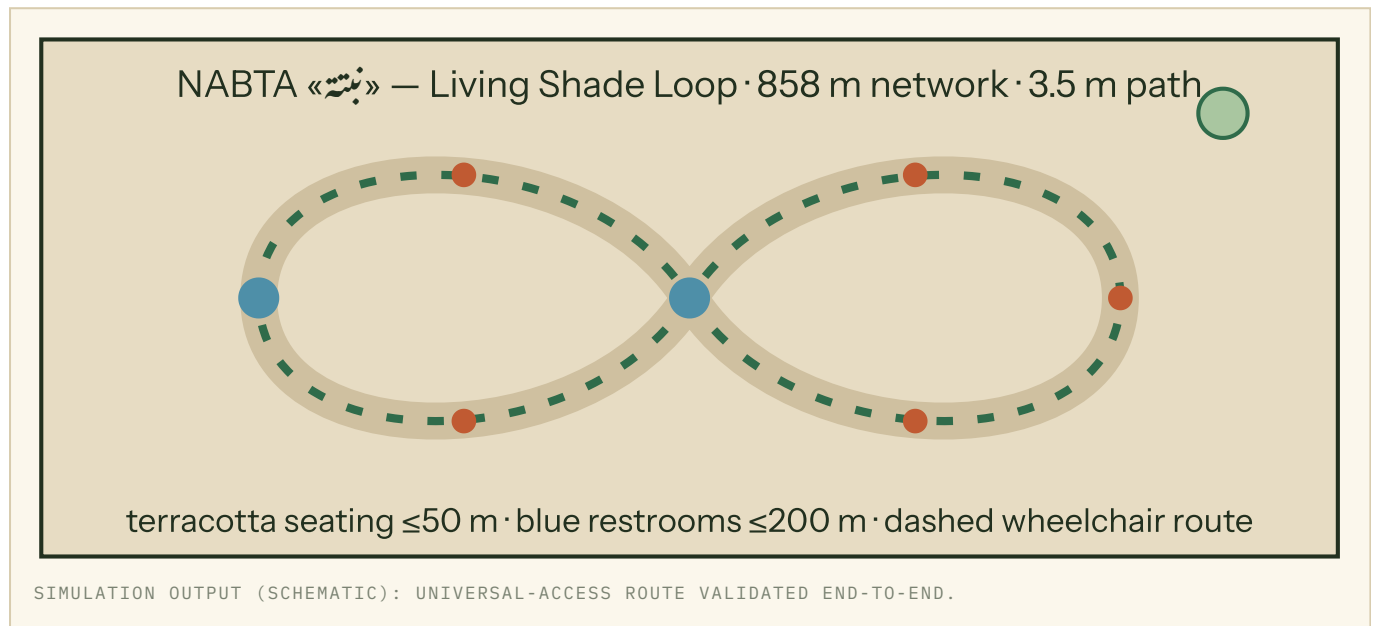
| DUBAI UNIVERSAL DESIGN<br>CODE THEME | NABTA MEASURE                                                                                                                                        |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Step-free movement                   | 100% step-free park: zero kerbs on the loop, flush terracotta bridges over the wadi (≤1:21), tactile-edged not stepped.                              |
| Wayfinding                           | Continuous tactile guidance strip on all 858 m of figure-eight and ring; tactile/high-contrast decision studs at all ten room junctions.             |
| Rest & comfort                       | Seating every ≤50 m with armrests and backrests; 50% in year-round shade; wheelchair pause space at each bench.                                      |
| Arrival                              | 4 accessible parking bays nearest the south entry (3.6 m wide, shaded, ≤30 m to gate), level drop-off with blister paving.                           |
| Sanitary provision                   | Changing Places restroom (12 m², ceiling hoist, adult changing bench) at the entry plaza, plus ambulant-accessible cubicles at both restroom blocks. |
| Sensory inclusion                    | Quiet room adjacent to the sensory wadi; induction loop at the majlis plaza; predictable acoustic zoning.                                            |
| Signage                              | Braille + pictogram + high-contrast signage in Arabic and English at all entrances and room thresholds, mounted 1.2–1.6 m.                           |
| Inclusive play                       | Play dune equipment with transfer platforms at 450 mm, wheelchair-accessible carousel, ground-level play at every cluster.                           |

All dimensions meet or exceed Code minimums; each row is auditable line-by-line.

### C. AI-ASSISTED CIRCULATION VALIDATION (HUMAN-LED)

The plan was stress-tested by **agent-based pedestrian simulation**: the ~1,800 daily visits (385 simultaneous peak) per scenario — calibrated wheelchair (0.8 m/s), stroller, elderly, child and jogger profiles — across Friday-peak, school-run and summer-evening demand. *AI verified, humans decided*:

- Confirmed no crossing point where jogger and wheelchair flows conflict at >0.3 encounters/min; this moved the terracotta jogger lane to the outer edge.
- Validated every wheelchair agent completes the network with seating ≤50 m apart and a restroom within 200 m — simulation-verified, not rules-of-thumb.
- Predicted desire lines matched the tertiary layer; two AI-suggested shortcuts were **rejected** on CPTED sightline grounds and logged in the decision record.



### D. ARRIVAL STRATEGY

- **Drop-off:** lay-by off the southern street edge per RTA geometry — 3 vehicle lengths, level boarding, tactile paving, shade canopy — sized for the school-run peak.
- **Parking:** ~40 car bays (incl. 4 accessible + 2 family) in a shaded grove lot; permeable paving.
- **Public transport:** signed, shaded, step-free links to 2 RTA bus stops within 300 m.
- **Micromobility:** 30 shaded bicycle racks per entrance (90 total) plus an e-scooter dock at each (south, west, east), outside pedestrian desire lines.

### E. SERVICE & EMERGENCY ACCESS

- Loop segments form a **4 m clear emergency route** (path + reinforced shoulder) giving Civil Defence and ambulance reach to all ten rooms; removable bollards at the three gates.
- **W-edge service yard:** screened compound with café back-of-house delivery, maintenance store, and irrigation plant; service vehicles never cross visitor space during opening hours.
- Six paired **waste/recycling points** on the loop, serviced from the W yard before 7:00 am.

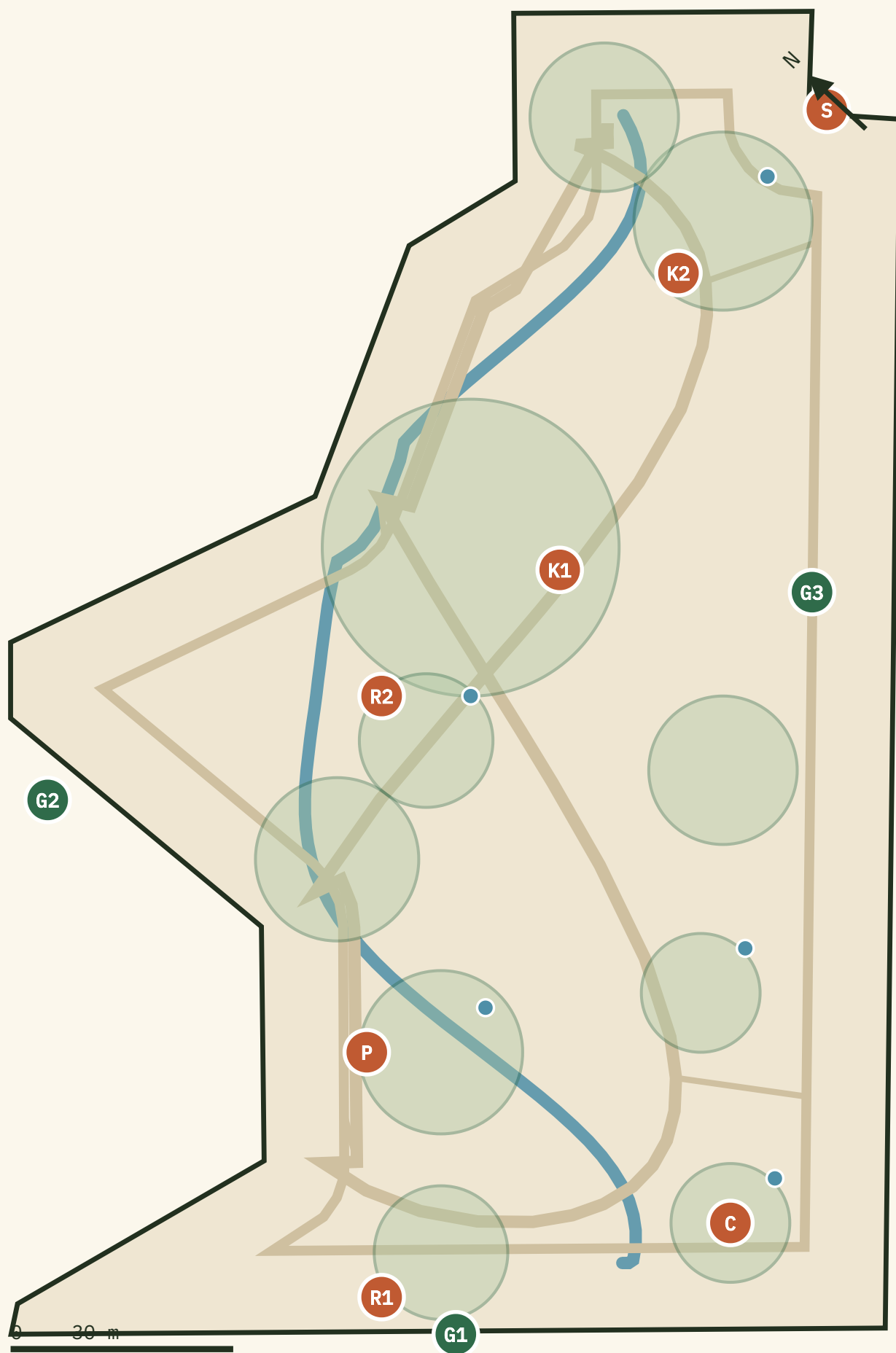
### F. INDICATIVE COMMERCIAL & SERVICE FACILITIES PLAN

**Catchment:** Al Safa 2 is a low-rise villa district — several thousand residents within a 10-minute walk, schools to the south, Sheikh Zayed Road offices minutes away. Demand peaks at dawn/evening, the school run and

winter weekends; fixed space is sized to base load, cart pads absorb peaks.

| FACILITY                             | LOCATION (GARDEN ROOM)   | CATCHMENT LOGIC                                                                          |
|--------------------------------------|--------------------------|------------------------------------------------------------------------------------------|
| Café + shaded terrace (120 m²)       | South entry plaza        | Captures school-run parents, loop finishers, and street footfall; visible from drop-off. |
| 2 kiosk cart pads (power/water)      | Event lawn edge          | Activated during events and weekends; zero fixed cost when idle.                         |
| Coffee/juice kiosk                   | Loop waist, by play dune | Serves caregivers with sightlines to all play zones.                                     |
| Restrooms ×2 (incl. Changing Places) | Entry plaza + loop waist | No point in the park >200 m from a restroom.                                             |
| Prayer room + wudu                   | Beside majlis plaza      | Adjacent to gathering space; on tactile route.                                           |
| Water refill + shade shelters        | Every second room        | Summer walkability; reduces single-use plastic.                                          |





**C** Café + terrace (concession)

**K1** Kiosk cart pads x2 – event lawn

K2 Coffee/juice kiosk – play dune

R2 Restrooms – loop waist

S Service yard – screened

G2 West gate

Water refill + shade shelter – every second room

R1 Restrooms + Changing Places

P Prayer + wudu – majlis

G1 Main gate – south

G3 East gate

INDICATIVE COMMERCIAL & SERVICE FACILITIES PLAN – PLOTTED ON THE SURVEYED 16-VERTEX PARCEL (OFFICIAL AS-BUILT DWG, 15,001 M<sup>2</sup>). COMMERCIAL FOOTPRINT <2% OF PARK AREA.

*Concession note:* café and kiosks are tendered as one concession, revenue share tied to mandated 06:00–23:00 opening — guaranteeing dawn and summer-evening service while the misted terrace keeps the south plaza usable; cart pads flex to winter events without permanent build-out. A wellness clause mandates a healthy-menu baseline and free water refills, keeping the commercial footprint under 2% of the 15,001 m<sup>2</sup> park.

#### G. SAFETY BY DESIGN

- **CPTED:** clear sightlines along the loop, understorey planting below 0.8 m at path edges, no dead-end paths, active café frontage at the main entrance.
- **Adaptive lighting:** loop 20 lux, garden rooms 10 lux, planted gardens 5 lux; all 2700 K warm white, full-cutoff dark-sky fixtures, presence-dimmed after 23:00.
- **Privacy commitment:** operational CCTV at entrances only; **no facial recognition** and no biometric analytics anywhere in the park.